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Modeling cryptocurrency failure using deep learning approaches and a post-hoc interpretability algorithm

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ABSTRACT

The lack of enforced trading termination in the cryptocurrency market allows for an exploration of the "natural death" of tradable assets. We propose two deep learning survival models to study this phenomenon and a post-hoc interpretability algorithm to interpret the results and test the hypothesis. The proposed deep learning survival models outperform the time-dependent Cox regression in both prediction performance and interpretation flexibility. Our results indicate that lower trading volume, market capitalization, social media attention, and higher price volatility predict increased cryptocurrency "death" hazards.

1. Introduction

Cryptocurrency technology has achieved remarkable success and attracted substantial developer engagement and investments. Cryptocurrencies are traded on specialized financial exchanges, but one noteworthy phenomenon in cryptocurrency exchanges, which is absent in the traditional stock exchanges, is the gradual cessation of trading (i.e., price and trading volume gradually fall to zero), signifying the "death" or failure of the cryptocurrencies.[1] Such trading discontinuation may occur during market downturns or when the related smart contract applications fail.

The failure of a cryptocurrency may seem comparable to the delisting events in the stock markets. However, there are notable differences. First, in a traditional stock exchange, listed companies must adhere to the standards set by the SEC and the exchanges; their shares will be delisted if these standards are not met [1]. However, due to the decentralized nature of cryptocurrency, regulatory authorities cannot effectively mandate a complete trading termination, although government enforcement actions may limit or discourage it. Therefore, the discontinuation of cryptocurrency trading is the result of a market equilibrium shaped by market dynamics and regulatory influences, rather than by direct regulatory orders. Second, in a traditional stock market, the decision to delist is usually publicly announced [2], allowing investors to make necessary adjustments. Conversely, in the cryptocurrency market, there is no formal "death notice," and investors do not receive direct warnings before a cryptocurrency's failure. Thus, they must make investment adjustments using their private information. Third, the price of failed cryptocurrencies will eventually fall to zero, but delisted stocks may still be traded at positive prices privately [3].

In the absence of direct trading termination orders, the cryptocurrency market presents a unique opportunity to examine the "natural death" of tradable assets resulting from market equilibrium. This paper investigates this phenomenon using various survival modeling approaches and compares their model interpretability and predictive performance. First, we estimate a Kaplan–Meier Survivor Function and a time-dependent Cox regression as the baseline. Second, we propose two deep learning survival models based on long short-term memory (LSTM) and transformer neural networks to address certain drawbacks of the Cox regression. Third, we propose a post-hoc interpretability algorithm to explore the lagged and nonlinear effects of the covariates using the trained deep learning survival models. The rationale of the interpretability algorithm is to compute the expected impulse response function and use it to show how perturbations that occur to certain covariates influence the "death" hazard over time. Last, we interpret the non-linear effect of the covariates on the next-day hazard using partial dependency plots of the deep learning survival models. In conclusion, our proposed deep learning survival models offer superior performance in predicting cryptocurrency "death" while also providing more comprehensive and

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[1] Please refer to the following examples of cryptocurrency failure: www.coingecko.com/research/publications/how-many-cryptocurrencies-failed. Accessed May 12, 2025.

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flexible model interpretations, especially regarding the lagged and nonlinear effects of the covariates.

This study makes several important contributions. First, in the traditional stock market, delisting decisions are typically governed by rigid and predefined regulations. Firms failing to meet certain financial and operational standards will be forced to delist from the exchange [4]. Therefore, examining the effect of the financial or operational performance on delisting is pointless, as the delisting decision is deterministic due to regulations. Indeed, the literature mainly focuses on the effect of non-financial/non-operational factors on delisting, such as industry categories [5], firm longevity [6], and firm size [7]. Taking advantage of the unique and underregulated cryptocurrency market, this research examines the association between cryptocurrencies' market dynamics and their failure, free from external regulatory enforcement to terminate the trading. As a result, our research manages to investigate the life cycle of a tradable asset only under market equilibrium, which is an unexplored research gap. We analyze the fundamental motivations behind trading behaviors through the lens of market microstructure theory and portfolio theory, and then apply their frameworks to develop hypotheses regarding effective predictors for the cryptocurrency death.

Second, we propose a deep learning survival modeling framework to address two important drawbacks of the traditional time-dependent Cox regression: (1) lagged effects of covariates on the hazard rate are not captured (Bharath and Dittmar 2010) and (2) it makes strong assumptions about the underlying stochastic process (e.g., the log of the hazard is a linear function of covariates). These restrictions negatively impact the model's performance and interpretation flexibility [8]. To address these issues, we developed two deep learning survival models based on the LSTM network and the transformer network and applied them to the survival data. These models are designed to predict the next-day death hazard using the complete historical time series of the predictors. Such models not only allow past observations to influence future hazard rate but also make no assumptions about the form of the underlying stochastic process (i.e., both the form of the stochastic process and the parameters depend on the covariates).

Last, this paper contributes to the interpretable deep learning literature by introducing an interpretability algorithm based on the impulse response technique to analyze the lagged effects of covariates in deep learning survival models. Although neural network models could approximate roughly any functions given sufficient complexity [9,10], interpreting the model and understanding its decision-making process is difficult for humans due to varying model structures and over-parameterization [11,12]. This study proposes an interpretability algorithm by emulating the neural network's data processing to create connections between inputs and outputs. Briefly speaking, we first introduce a small fixed perturbation (10 % of the standard deviation) to each covariate, one at a time, and then calculate and summarize its effect on the death hazard rate in the following days to assess the influence of each covariate. Using this procedure, we managed to compute the expected impulse response function (hereinafter IRF) for the deep learning survival models to evaluate how cryptocurrency death hazard responds to market shocks over time. This is the first time the impulse response technique is applied to survival analysis to produce model interpretations spanning a delayed period.

In a related effort, we also created partial dependency plots for the deep learning survival models to examine the nonlinear effects of covariates. We introduce varying perturbations to each covariate and visualize their average impact on the next-day death hazard. Traditionally, a researcher may only assess the linear effects of the covariates using the Cox model. However, the actual effects could be non-linear (e.g., concave increasing, convex increasing, bell-shaped), and relying on a generalized linear model may oversimplify the relationships and overlook important details. Our interpretability approach shows the potential of using deep learning models for hypothesis testing, an area typically dominated by interpretable linear models.

The rest of the paper proceeds as follows. Section 2 reviews the related literature and develops hypotheses. Section 3 describes the data collection procedures. Section 4 discusses the traditional and proposed survival analysis modeling techniques. Section 5 presents the prediction performance comparisons. Section 6 illustrates the proposed IRF-based interpretability algorithm. Finally, the paper concludes in Section 7.

2. Literature review and hypothesis development

2.1. Survival analysis in business analytics and its integration with deep learning

Cox regression is a statistical technique used to model the time until an event occurs [13]. It is mainly used in medical and epidemiology research to understand the relationship between a set of covariates and the time until a particular event happens, such as death and disease recurrence [14].

However, Cox regression has been increasingly used in social sciences. For example, Boršič and Kavkler use it to study the factors influencing the unemployment duration and find that higher education levels reduce the time to find jobs [15]. Agarwal explores the survival of new entrants into an industry and finds that entrants' survival rate declines with increased competitive intensity [16]. Sanizah et al. study the time to divorce decisions and find that the age at marriage, presence of children, couple's educational level, employment status, and household income are significant predictors [17].

In a more relevant context, Cox regressions are also used in the finance literature to investigate the occurrence of significant financial events. For example, Eilnaz and Meziane study the decision of voluntary delisting and report that firms with high leverage, high insider ownership, and low profitability/growth opportunity/trading volume are more likely to voluntarily delist from stock exchanges [18]. Shumway uses Cox regression to forecast bankruptcy and finds that variables such as firms' market size, past stock returns, and idiosyncratic returns are important predictors. In contrast, many accounting-related variables are statistically unrelated to bankruptcy probability [19]. Maria and Lyn propose a Cox model-based credit scoring system that predicts early repayment and default for personal loans and demonstrates a competitive performance when compared to the industry standard classification approaches [20].

Although Cox models are widely used in survival analysis across various fields, there are several noted limitations. First, because the Cox model does not have an inherent mechanism to capture inter-period dependencies, investigating the lagged effects of the covariates is difficult unless certain lagged or cumulative covariates are included in the model estimation [21]. Second, even though the inclusion of the lagged or cumulative covariates into the models may address the first issue, these covariates must be custom-designed by researchers and can be rather subjective. For example, studies may use distinct time lags and methodologies to compute the cumulative covariates, and the appropriateness of these measurements relies significantly on the researchers' domain knowledge [22]. Third, the model makes strong assumptions regarding the form of the underlying stochastic process (i.e., the logarithm of hazard is a linear combination of covariates), and nonlinearity is generally not considered in the model [8]. The second and third issues have become more prominent with the advancement of deep learning technologies. In this research, we propose a deep learning survival modeling framework that potentially addresses all three issues. These models employ LSTM and transformer neural network structures to inherently capture the inter-period effects of covariates [23] while eliminating the need for manual feature engineering and assumptions for the underlying stochastic process [24].

Although there are several studies exploring deep learning-based survival analysis, each of them falls short in certain considerations [4–6], for example: (1) the hazard rate or the outcome variable at time t depends on the covariates observed during the same time t period, it is thus less practical to use the model in prediction; (2) the model is not a

time-dependent survival model; (3) there is no interpretability strategy proposed to obtain managerial insights; (4) there is no performance comparison between the deep learning survival model and the Cox regression; and (5) the loss function used in model training is not regularized to focus on "death" event prediction, limiting the model's performance. Details will be presented in later sections.

2.2. Interpretability algorithms for deep learning model

Deep learning algorithms started dominating prediction benchmarks around 2012 in many domains such as banking, retailing, healthcare, recommendation systems, and natural language processing [25]. In addition, deep learning allows machines to automatically discover relevant features and learn data representations that would otherwise be laborious and complex tasks for humans [11]. Such capability is achieved through increasingly complex hierarchies and over-parameterization. However, one major drawback is that such models' rationale and decision-making process are usually unclear, reducing their capability to interpret and explain the underlying mechanisms [26].

The lack of interpretability hinders the adoption of deep learning models because they are difficult to trust, especially in high-stakes applications such as healthcare and financial services [27,28]. Research has found that deep learning models can be easily "fooled" to misclassify and generate unpredictable outputs in image processing [29] and natural language processing [30]. These undesirable behaviors sparked many recent studies to develop interpretability algorithms to gain users' trust and produce insights about the model's decision-making process [31].

The type of interpretability algorithm that is most relevant to this particular research is commonly known as the post-hoc interpretability algorithm, in which a separate explaining model is created to explain the main model [11]. There are many existing interpretability algorithms falling under this category. For example, Mukund et al. proposed the "Integrated Gradient" method, which attributes the contribution of each input feature to a neural network's output by integrating gradients along a path from a baseline to the input [32]. Avanti et al. proposed the "DeepLIFT" method, which determines the contribution of each unit based on the difference between its activation value and the reference activation, whereas the reference activation is computed using all units during forward propagation [33]. Marco et al. proposed the "LIME" algorithm (local interpretable model-agnostic explanations). This methodology randomly samples data around inputs, computes their predictions using the black box model, and then trains a separate interpretable model on this perturbed dataset. The black box model is thus explained using the interpretable local model [34]. Scott and Su-in proposed the "SHAP" algorithm (Shapley Additive explanations), which assigns each feature an importance value for a model's prediction by computing fair contribution scores based on cooperative game theory [35].

The proposed interpretability algorithm is created specifically to capture the lagged effect of covariates for longitudinal analysis. For survival analysis in particular, this method aims to estimate an IRF that depicts how perturbations to a covariate influence the hazard rate over time. The perturbations are introduced to each covariate separately, and their influence on the hazard rate is summarized to generate interpretations and insights.

2.3. Delisting from exchange

Delisting from the exchange primarily occurs in stock markets and refers to the removal of a company's stock from an exchange [1]. There are two types of delisting: involuntary delisting and voluntary delisting [3]. Involuntary delisting occurs when a company is forced to delist from an exchange because of an SEC rule violation, failure to meet the exchange's requirements, merger and acquisition, bankruptcy, or migration to another exchange [36]. Voluntary delisting occurs when a company withdraws its stock listings by repurchasing its shares or going to over-the-counter

Market [37]. For the two delisting types, research focuses are drastically different. Since the reasons for involuntary delisting are usually deterministic (i.e., violations of government or exchange regulations), most studies investigate the consequences of involuntary delisting, which are mainly equity value decline, decreased liquidity, and lower trading volume [1,2,38,39]. In contrast, research on voluntary delisting primarily focuses on its reasons. According to the literature, companies voluntarily delist from stock exchanges mainly due to the inability to obtain enough funding to rebalance their leverage, limited growth opportunities, significant insider control, and high compliance costs (i.e., audit fees) [4,18].

However, no matter how a stock is delisted, it is either due to regulatory requirements or a decision made by management. But what does the "natural lifecycle" of a tradable asset look like? How does a tradable asset reach its "natural death" without enforcement? To the best of our knowledge, there is currently no answer in the existing literature. The cryptocurrency market offers an ideal research setting to address these questions because enforced trading termination is practically impossible for cryptocurrencies due to their decentralized nature. This research aims to investigate the predictive factors for cryptocurrency's "natural death" using survival analysis to shed light on this unexplored area of research.

2.4. Hypothesis development

In this section, we draw insights from existing literature to identify four predictors for cryptocurrency failure. They include trading volume, market capitalization, social media attention, and price volatility.

To study why trading stops, we must first understand the motivations behind stock trading. According to market microstructure theory, the most prominent reasons for trading are the heterogeneous expectations about asset values [40]. Private information and asymmetric information lead to investors' unique insights and market evaluations, contributing to price discovery. Even in an efficient market without private information (e.g., all information is freely available), differences in information interpretation also create heterogeneous beliefs that motivate trades [41]. While according to the portfolio theory, investors trade to rebalance and adjust their assets in response to changes in wealth, risk tolerance, consumption needs, liquidity needs, and investment opportunities [42], which lead to trading even in the absence of new market information. These theoretical frameworks provide a strong foundation for understanding trading halts in the cryptocurrency market, and with which we will develop our hypotheses.

The daily trading volume of a cryptocurrency refers to the amount of cryptocurrency exchanged during that day. A significant decrease in trading volume has several implications. First, it suggests that the investors barely have heterogeneous evaluations, indicating that the market may have already reached a consensus on the future of the cryptocurrency (i.e., death), which significantly reduces the number of interested buyers on the market. Second, a sharp decline in trading volume also suggests that the buying and selling of the cryptocurrency no longer address investors' portfolio rebalancing needs. This implies that investors see limited investment opportunities and do not expect the trading of the cryptocurrency to meet their consumption needs, liquidity needs, and risk adjustment goals, indicating that its value has already fallen significantly. Therefore, we propose a negative relationship between trading volume and cryptocurrency death hazard.

Hypothesis 1. *decreased trading volume predicts increased cryptocurrency death hazard.*

The business failure literature has demonstrated that higher market capitalization correlates with a lower probability of failure [5,7,43]. Yet, it remains unknown whether this holds for tradable assets as well. As

discussed previously, investors have idiosyncratic investment needs and portfolio rebalancing needs [44]. A cryptocurrency market with a large market capitalization is expected to absorb these needs more effectively without affecting the stability and confidence in the market [45]. Conversely, in a low market capitalization market, large selling orders are more likely to create order imbalances, which are costly to absorb by the market makers. Such order imbalance may lead to large bid-ask spreads, significantly reduced prices, and even panic selling [46]. Therefore, we propose a negative relationship between market capitalization and cryptocurrency hazard.

Hypothesis 2. *decreased market capitalization predicts increased cryptocurrency death hazard.*

The emergence of social media renovated retail investors' access to new information [47]. Previously, investors mainly relied on public disclosures such as quarterly earnings releases, analyst reports, and market news to aid their decision-making. As social media becomes prevalent, retail investors increasingly use social media messages to make trading decisions [47]. Social media users are motivated to share information with others due to many factors such as reputation, tenure in the field, and reciprocity [48]. Besides these non-economic motivations, economic motivations also exist, especially in investment communities. For example, information shared by a message poster could influence the viewers' trading decisions and expedite the convergence of market prices to the message poster's expectation [47].

In light of these arguments, we expect social media attention to be an effective predictor for cryptocurrency failure. If a cryptocurrency receives minimal social media attention, it signifies a lack of investor interest. Sharing information about a diminishing cryptocurrency fails to contribute to the goals of improving reputation, establishing tenure in the field, or impressing peers. Therefore, social media users will be reluctant to post messages about diminishing cryptocurrencies, and we thus propose the third hypothesis.

Hypothesis 3. *decreased social media attention predicts an increased cryptocurrency death hazard.*

Most cryptocurrencies are peer-to-peer cash/currency systems. But unlike fiat currencies, cryptocurrencies are usually characterized by high volatility [49], indicating that speculation plays a major role in the cryptocurrency market rather than its application in trade settlement. For a currency to function effectively as a medium of exchange and a reliable store of value, it needs to be stable and non-volatile [50]. While some businesses, such as Apple and Best Buy, have started accepting cryptocurrency as payment for their products and services, the high volatility and the associated risks of cryptocurrencies significantly limit their adoption by small businesses [51].

Excessive volatility is also undesirable for an investment asset because investors are presumably risk-averse and dislike volatility [52]. Studies have indicated that the risk premium of volatility tends to be mostly negative across various contexts [52,53]. According to the portfolio theory, for a portfolio to remain on the "efficient frontier," high volatility must be compensated with high returns [54]. If a highly volatile cryptocurrency fails to outperform others in returns, investors will exchange it for other investment opportunities to optimize their portfolio, negatively influencing the prospect of the cryptocurrency and increasing the hazard.

In summary, we argue that high volatility negatively influences the potential of a cryptocurrency as a currency and as an investment asset and propose the next hypothesis.

Hypothesis 4. *increased price volatility predicts increased cryptocurrency death hazard.*

3. Data

3.1. Cryptocurrency trading data

We downloaded the daily cryptocurrency trading data of 8194 cryptocurrencies, including prices, trading volumes, and market capitalization from July 13, 2010, to June 30, 2024, from CoinMarketCap. We set the beginning of the sample period to be July 13, 2010, because it is the first day that CoinMarketCap provides data about Bitcoin. CoinMarketCap is a website specializing in cryptocurrency trading data such as prices, trading volume, and market capitalization. According to SimilarWeb, a global website traffic analytics platform, coinmarketcap.com is ranked 865 globally in terms of website traffic (as of April 2025). CoinMarketCap is also the data source for many other cryptocurrency-related studies [55,56].

According to the classic survival analysis framework, a cryptocurrency either survived beyond the data collection period (i.e., censored by the dataset) or "died" before it. Given the lack of a formal definition in the literature, we define cryptocurrency failure as trading discontinuation. Formally, we consider a cryptocurrency failed or "dead" on day t_0 if the trading volume is zero across t_0, t_1, and t_2. A failed cryptocurrency may resume trading in rare cases, but we consider only the first failure, to be consistent with the traditional survival analysis.

Among all cryptocurrencies in the sample, some are barely traded and "die" too soon. We exclude these short-lived cryptocurrencies because cryptocurrency failure could be attributed to an unsuccessful Initial Coin Offering (i.e., unsuccessful project launching), but not to the factors of interest in this study. The influence of Initial Coin Offering is more likely to be factored into the price of those stable cryptocurrencies that have already been in existence for some time. After the screening, 2701 cryptocurrencies that survived >60 days remained in the sample. And their daily return, market capitalization, trading volume, and volatility are obtained.

3.2. Cryptocurrency social media data and search data

To capture each cryptocurrency's social media attention, we obtain social media data from Bitcointalk.org, one of the most popular online communities for cryptocurrency investors. Many cryptocurrency studies have used this online community as the primary data source [47,57,58]. This online community provides a message board dedicated to "altcoin" social media discussions (altcoin refers to cryptocurrencies other than Bitcoin), and we use this message board as the data source to obtain social media data. We matched each cryptocurrency with its social media messages using the appearance of its ticker symbol in the messages.

Besides social media, we use Google Trend data to control for popularity. Specifically, we use the cryptocurrency's ticker symbol as the Google Trend search terms and obtain the monthly Google Trend score over each cryptocurrency's life cycle to capture their popularity and attention over time.

4. Survival analysis modeling

In this section, we first discuss conventional survival modeling approaches such as the Kaplan–Meier estimator and the Cox regression and then propose a deep learning survival modeling framework.

4.1. Survivor function estimation using Kaplan – Meier estimator

Kaplan–Meier survivor function estimator (KM estimator) is a nonparametric statistical method to estimate the survivor function [21]. We observe each cryptocurrency i ($i = 1 \ldots n$) during each day t ($t = 1, 2, 3, \ldots, T$, where T is the last observation date). Following the standard survival analysis literature, if cryptocurrency i dies before T, we use t_f to

denote its failure time. If cryptocurrency i's failure does not occur before T (i.e., surviving beyond the last observation date), then it is censored [59]. Each cryptocurrency i is either dead or censored at the end of the data collection period (June 30, 2024). Consistent with the literature, we define the survivor function as $S(t) = \Pr\left(t_f > t\right)$ [21], which reflects the probability that a cryptocurrency survives beyond t. The KM estimator is simply an estimator of the survivor function, expressed as:

$$\widehat{S}(t) = \prod_{t_1 < t_j < t} \left(1 - \frac{f_j}{n_j}\right), \tag{1}$$

where $t_1 < t_2 < \ldots < t$ are unique days on which at least one death event occurred. During day t_j when there is at least one death event, denote f_j as the number of death events occurred and n_j as the number of cryptocurrencies at risk (i.e., the number of cryptocurrencies that neither died nor were censored).

We estimate the KM survivor functions using different subsamples to provide preliminary support for our hypotheses. These subsamples include (1) cryptocurrencies with above/below the median of the average trading volume; (2) cryptocurrencies with above/below the median of the average market capitalization; (3) cryptocurrencies receiving above/below the median total number of social media mentions; and (4) cryptocurrencies with above/below the median of the price volatility. Fig. 1 illustrates these estimated KM survivor functions, and the results suggest that cryptocurrencies traded more actively, capturing a larger market capitalization, receiving more social media attention, and having a lower price volatility are associated with lower death hazards. These results support all four hypotheses.

4.2. Time-Dependent Cox survival regression

The traditional Cox proportional hazard model assumes that the hazard at time t is an exponential function of a linear combination of the subjects' time-invariant covariates, as shown:

$$h(t) = h_0(t) \times \exp\left(\beta X\right), \tag{2}$$

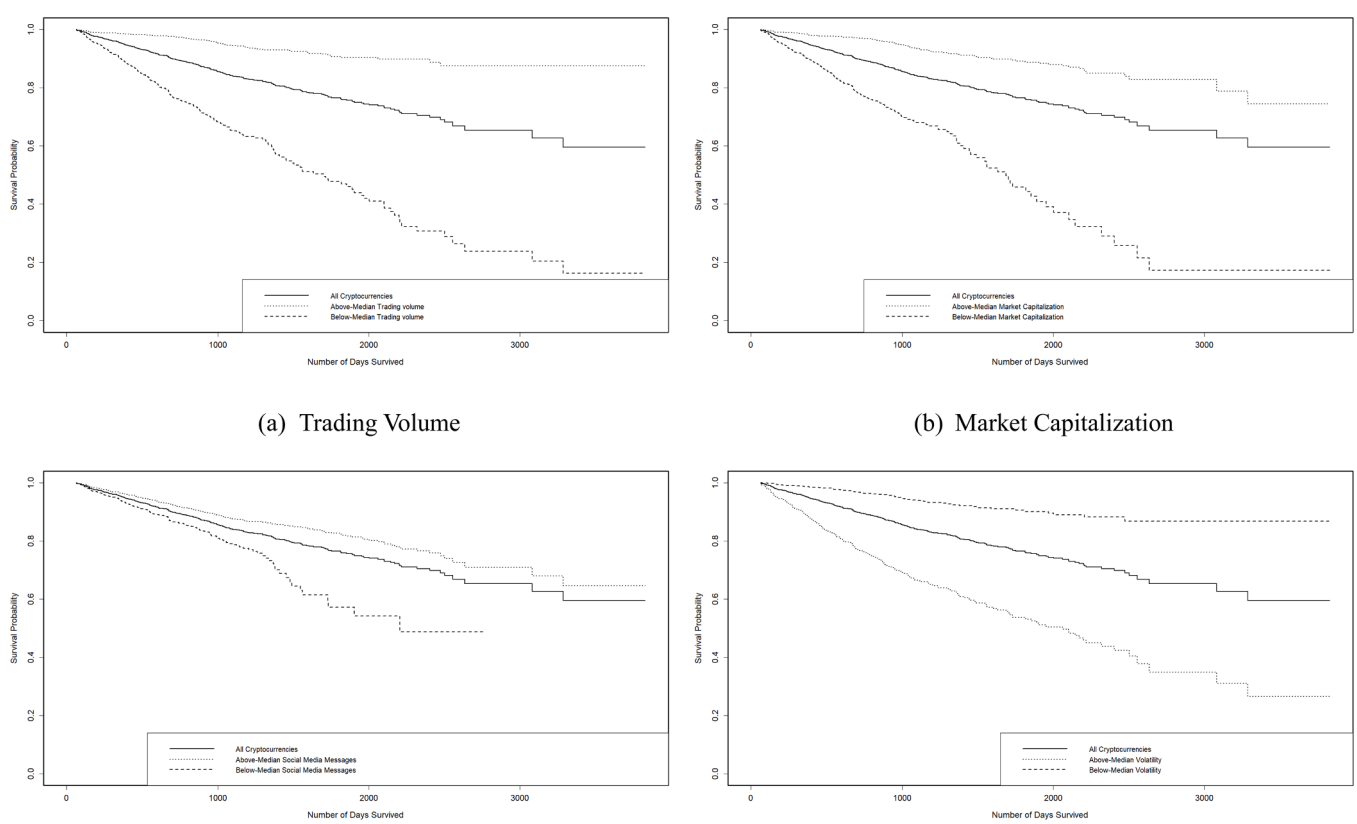
where $h(t)$ is the time t hazard rate, reflecting the instantaneous probability of death at time t conditional on surviving through time $t-1$, $h_0(t)$ is the baseline hazard at time t, and X are time-invariant covariates for the subjects. An important restriction of this model is that the researchers do not observe the covariates X in continuous time, and they must carry the last observed values of the covariates into future hazard computations.

Relaxation of this restriction leads to the time-dependent Cox regression, where the hazard depends on longitudinal covariates. We follow the literature to analyze cryptocurrency death with the following time-dependent Cox regression [18,60,61]:

$$h(t+1 \mid X(t)) = h_0(t+1) \times \exp\left(\beta X(t)\right), \tag{3}$$

where $h(t+1 \mid X(t))$ is the time $t+1$ hazard rate for a cryptocurrency with time t covariates $X(t)$, $h_0(t+1)$ is the baseline hazard at time $t+1$, the term $\exp\left(\beta X(t)\right)$ is the hazard rate deviation from the baseline hazard due to $X(t)$, and $\exp\left(\beta\right)$ is the hazard ratio indicating the changes in hazard given a unit change in the covariates. A hazard ratio greater than (less than) 1 indicates that a particular covariate is positively (negatively) related to the hazard rate. Note that we predict the death hazard of time $t+1$ using time t covariates to ensure a properly designed predictive model, and this type of model configuration is in line with the literature [47,61,62].

Although time-dependent Cox regression allows for a longitudinal



(a) Trading Volume (b) Market Capitalization

(c) Social Media Messages (d) Volatility

Fig. 1. Kaplan–Meier Survivor Function Estimation for Subsamples.

survival analysis, the covariates observed at time t only influence the time $t+1$ hazard by its design. However, lagged effects are not directly modeled by time-dependent Cox regression because it does not offer any mechanisms to capture such inter-period dependencies. Therefore, we calculate a series of moving average measurements over a previous period to account for the trend and lagged effects. We compute two groups of variables. The first comprises the trading dynamics of cryptocurrencies, including (1) $Return_t$, which is the cumulative returns over the past 30 days; (2) $TradingVolume_t$ and $MarketCap_t$, which are the 30-day moving average trading volume and market capitalization, respectively; (3) $VolumeStdev_t$, which is the standard deviation of the trading volume over the past 30 days; (4) $NumHighReturn_t$ and $NumLowReturn_t$, which are the number of days in the past 30 days that experience a return larger than 10 % or lower than −10 %, respectively; (5) $Volatility_t$, which is the sum of squared returns over the past 30 days; and (6) $ReturnSlope_t$, $CapSlope_t$, $VolumeSlope_t$, and $VolatilitySlope_t$, which are the slope obtained by regressing $Return_t$, $MarketCap_t$, $TradingVolume_t$, and $Volatility_t$ on t using the past 30 days' data, respectively, thereby capturing the general short-term trend of these variables.

The second group of variables consists of social media attention received by the cryptocurrencies, including (1) $AnyMention_t$, which is a binary variable set to 1 if there is at least one social media mention during the past 30 days; (2) $NumMention_t$, which is the number of social media mentions during the past 30 days (set to 0 if $AnyMention_t = 0$); (3) $NegSentiment_t$, which is the total number of bearish words divided by the total number of words in all messages mentioning a particular cryptocurrency during the past 30 days (set to 0 if $AnyMention_t = 0$) [63]; and (4) $DaysSinceLastMention_t$, which is the number of days since the cryptocurrency was last mentioned in any social media messages.

Apart from the 30-day moving average measurements, we also calculate another set of similar variables using cumulative aggregation (i.e., variables computed using a time window spanning from $t = 1$ to the current day) to explore the effect of the long-term trend. Under cumulative aggregation, $NumHighReturn_t$ and $NumLowReturn_t$ are replaced by $PctHighReturn_t$ and $PctLowReturn_t$, which measure the percentage of trading days with a return greater (less) than 10 % (−10 %). Last is $GoogleTrend_t$, which is the monthly observed Google Trend data for each cryptocurrency. Table 1 shows the summary statistics of these variables.

We present the model estimation results in Table 2, which shows the model estimated using 30-day moving average variables in columns (1) and (2) and the model estimated using cumulative variables in columns (3) and (4). The result shows that trading volume is strongly associated with the hazard rate. The coefficient estimates for TradingVolume are consistently negative and statistically significant for both models, indicating that a 1 % increase in the 30-day moving average trading volume (cumulative average trading volume) is associated with a 0.87 % (0.46

Table 2
Time-Dependent Cox Survival Regression Estimation.

	30-day Moving Average		Cumulative	
	Coefficient (1)	Hazard (2)	Coefficient (3)	Hazard (4)
DaysSinceLastMention	0.0001* (1.739)	1.0001*	0.0002*** (3.185)	1.0002***
AnyMention	0.077 (0.357)	1.080	−0.05 (−0.425)	0.951
log (NumMention + 1)	−0.104 (−0.732)	0.901	−0.065** (−2.213)	0.937**
NegSentiment	−24.720** (−2.009)	1.841e-11**	−10.270** (−2.099)	3.471e-05**
Return	−0.005 (−0.421)	0.995	−0.005 (−0.094)	0.996
log (MarketCap + 1)	−0.004 (−0.222)	0.996	−0.061** (−2.401)	0.941**
log (TradingVolume + 1)	−0.870*** (−16.298)	0.419***	−0.464*** (−7.132)	0.629***
log (VolumeStdev + 1)	0.570*** (10.380)	1.768***	0.210*** (3.375)	1.234***
Volatility	−2.164e-07 (−0.632)	1	1.455e-07 (1.014)	1
NumHighReturn	0.054*** (2.883)	1.055***		
NumLowReturn	0.048*** (2.674)	1.049***		
PctHighReturn			−0.258 (−0.193)	0.773
PctLowReturn			5.291*** (4.309)	19.85***
ReturnSlope	0.227 (0.769)	1.254	2.562 (1.484)	12.96
CapSlope	0.224 1.346** (2.186)	3.844**	2.253 (0.486)	9.518
VolumeSlope	−10.050*** (−10.928)	4.314e-05***	−41.14*** (−5.348)	1.360e-18***
VolatilitySlope	−4.398e-09 (−0.002)	1	6.273e-07 (0.160)	1
GoogleTrend	−0.002 (−1.372)	0.998	−0.0007 (−0.544)	1
# Obs.	2,345,121		2,345,121	
Likelihood Ratio Test	Likelihood ratio = 2401; df = 16; $p = 0$		Likelihood ratio = 1474; df = 16; $p = 0$	

Note: ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table 1
Descriptive Statistics.

	30-day Moving Average					Cumulative Aggregation				
	Max	Min	Median	Mean	Std.Dev	Max	Min	Median	Mean	Std.Dev
DaysSinceLastMention	3,935	0	98	333.600	531.554	Same as the values in the first column				
AnyMention	1	0	0	0.189	0.391	1	0	1	0.814	0.390
NumMention	6,499	0	0	4.978	71.871	110,285	0	10	347.345	3,258.147
NegSentiment	0.156	0	0	0.002	0.006	0.154	0	0.006	0.007	0.007
Return	1,822.092	−0.138	0.0002	0.038	5.708	71.921	−0.140	0.003	0.019	0.675
MarketCap	5.259e+11	0	8,749,016	5.143e+08	7.392e+9	1.214e+11	193.632	1.498e+07	2.749e+08	2.361e+09
TradingVolume	1.581e+11	1	373,738.7	8.954e+07	1.758e+09	2.908e+10	5.166667	1,011,067	4.684e+07	6.308e+08
VolumeStdev	1.119e+11	0	176,020.4	3.637e+07	7.030e+08	3.711e+10	6.356	1,504,077	8.149e+07	9.477e+08
Volatility	7.855e+08	0	0	10,451	2,467,021	2.267e+07	0.0002	0.142	2,462.234	138,040.3
NumHighReturn	18	0	2	2.386	2.587					
NumLowReturn	21	0	1	2.212	2.557					
PctHighReturn						0.500	0	0.083	0.095	0.058
PctLowReturn						0.710	0	0.074	0.091	0.063
Return_Slope	178.656	−184.749	0	0.00002	0.459	0.955	−6.526	0	0	0.014
CapSlope	0.371	−0.487	−0.025	−0.013	0.073	0.155	−0.109	−0.001	−0.002	0.013
VolumeSlope	0.233	−0.428	−0.007	−0.006	0.046	0.137	−0.171	−0.001	−0.002	0.009
VolatilitySlope	1.228e+07	−1.238e+07	−0.0001	0.404	2.846e+07	1,599,947	−61,183.460	0	13.231	4313.305
GoogleTrend(monthly)	100	0	55	51.15	26.948	Same as the values in the first column				

%) decrease in the next-day hazard rate, supporting *Hypothesis 1*. However, high trading volume fluctuation tends to predict increased death hazards. Specifically, a 1 % increase in 30-day trading volume standard deviation (cumulative trading volume standard deviation) is associated with a 0.57 % (0.21 %) increase in the next-day death hazard.

The coefficient estimate for market capitalization is statistically significant in the cumulative variable model, indicating that a 1 % increase in the cumulative market capitalization is associated with a 0.06 % decrease in the next-day death hazard. These results are consistent with the KM estimator analysis and support *Hypothesis 2*.

The coefficient estimates for $DaysSinceLastMention_t$ in both models range from 0.0001 to 0.0002 and are statistically significant, meaning that on average, one more day without a social media mention is associated with a 0.1 % to 0.2 % increase in the next-day death hazard. This evidence supports *Hypothesis 3*. However, the coefficients for volatility are not statistically significant in both models and thus *Hypothesis 4* is not supported by the Cox regression.

4.3. Deep learning survival models

Section 2.1 describes several drawbacks of the conventional Cox survival models. In this section, we propose a deep learning survival modeling framework to address these issues.

The dataset used in this study contains daily observations of the covariates and the status variable (i.e., "alive" or "dead") over the entire lifecycle of each cryptocurrency, which resembles a time series dataset and can be modeled using certain types of neural networks, such as LSTM and transformer, because they are capable of capturing inter-period dependencies, and thus the lagged effects of the covariates [64]. The LSTM network achieves this purpose by forwarding memory vectors from the past to the future [65]; the transformer network does this using a matrix reflecting the dependency among different days [66]. Furthermore, deep learning approaches eliminate assumptions about the underlying stochastic process and the need for subjective feature engineering. According to the universal approximation theorem, deep learning models could approximate any function given enough structural complexity and proper non-linear transformations [67]. They are also recognized for their capability to automatically identify relevant feature representations for accurate predictions [24].

Several considerations must be taken when using LSTM/transformer networks for the purpose of survival modeling. First, after the death of a cryptocurrency, all covariates are no longer observable. To enable batch training, padding the sequence of the dead cryptocurrencies is necessary to match the length of the surviving ones, both for the predictors and for the status variable. Therefore, in addition to the "alive" status and the "death" status, we introduce a filler status "post-death." When training the transformer survival model, we apply a mask to block out the post-death dates so that the network will not learn dependencies for these post-death dates [66]. Also important is that the loss associated with the filler status "post-death" must not affect the cross-entropy loss function during training, so the model is forced to only improve the "alive" and "death" prediction accuracy.

Second, in our cryptocurrency survival analysis, there is only one "death" event at most (although in rare cases, a cryptocurrency may be "resurrected" after its "death," we only model the first "death"), but many "alive" days before it. Therefore, equally weighting different days in the total loss calculation will result in underperformance in death prediction because the loss contributed from the single death event among all days is negligible. To regularize the network to put more emphasis on "death" prediction instead of "alive" prediction, we introduce a hyperparameter λ to amplify the loss from the death event prediction.

Last, the effect of the technology lifecycle of each cryptocurrency should be considered [68]. The death hazard in different lifecycle stages is likely to differ. We use positional embedding for this purpose. We give each day a fixed vector to control for the time effect. Consistent with the literature, we compute the positional embeddings using the following equations [66]:

$$p(t,2i) = \sin\left(\frac{t}{n^{2i/d}}\right);\ p(t,2i+1) = \cos\left(\frac{t}{n^{2i/d}}\right). \tag{4}$$

In these equations, t indexes the days; $2i$ and $2i+1$ index the even and odd positions along the embedding vectors, respectively; d is the length of the embedding vector with $0 \le i < d/2$; and n is a user-defined scalar, which we set to 10,000. After incorporating these considerations, the modified cross-entropy loss function is as follows:

$$L = -\frac{1}{m}\sum_{i=1}^{m}\sum_{t=1}^{T}\left\{\lambda I_{S_i^t=death}\log\left(p_i^t\right)\right) + I_{S_i^t=alive}\log\left(q_i^t\right)\}, \tag{5}$$

where p_i^t is the predicted day $t+1$ death hazard of cryptocurrency i using day t covariates, q_i^t is the predicted day $t+1$ probability that cryptocurrency i survived, S_i^t is the true state of cryptocurrency i during day $t+1$ (i.e., "alive," "death," or "post-death"), and m is the sample size. Grid search is employed to identify the optimal hyperparameters for the model and the training process. These include λ (ranging from 500 to 1500, in steps of 100), and the number of LSTM/transformer layers in the model (from 1 to 3). All models are trained for a maximum of 200 epochs, and early stopping is implemented to prevent overfitting. Training stops if the test set loss does not improve for 20 consecutive training epochs. In addition, the learning rate is scheduled to be halved if there is no loss improvement in 10 consecutive training epochs. Because the scale of the variables used in prediction varies dramatically, we also normalized each feature using all pre-censored/pre-death data to ensure consistent variable scales and improved numerical stability. Both the training and testing datasets are normalized against the training dataset to ensure that the testing dataset is completely unseen during training.

Fig. 2 depicts the neural network. First, we concatenate the input vector each day with the positional embedding; second, we forward the concatenated input into one or more LSTM layers or transformer self-attention layers; third, we further process the outputs using fully connected feed forward networks; and fourth, we apply Softmax transformation to generate the predicted next-day death hazard. To be consistent with the Cox regression estimation, we use day t covariates to predict the day $t+1$ hazard rate.

Unlike the Cox survival regression, LSTM/transformer networks can autonomously extract valuable features and grasp dependencies between different days [24]. Therefore, there is no need to design lagged variables or cumulative variables manually, the model directly takes the normalized daily observed covariates as inputs. However, variables such as $DaysSinceLastMention_t$, $AnyMention_t$, and $GoogleTrend_t$ are included directly as inputs because they cannot be disaggregated into daily observed data. The interpretation of these models is presented in Section 6 after we propose the interpretability algorithm.

5. Cryptocurrency failure prediction performance evaluation

To gauge the out-of-sample hazard rate prediction performance, we use cryptocurrencies listed before September 15, 2020, on CoinMarketCap as the training set to train the models (both the Cox survival regression and deep-learning survival models) and use all cryptocurrencies listed after September 15, 2020, as the testing set to evaluate the prediction performance. In this way, we adhere to the existing literature to train the models with historical data and use them to forecast future outcomes. This allows us to reliably assess their prediction performance and generalizability in future applications [5]. We use September 15, 2020, to split the sample into the training set and the testing set because approximately 20 % of the death events (186 death events) occurred after September 15, 2020, and about 80 % (740) occurred before it. This results in an approximate 80 %/20 % train-test split.

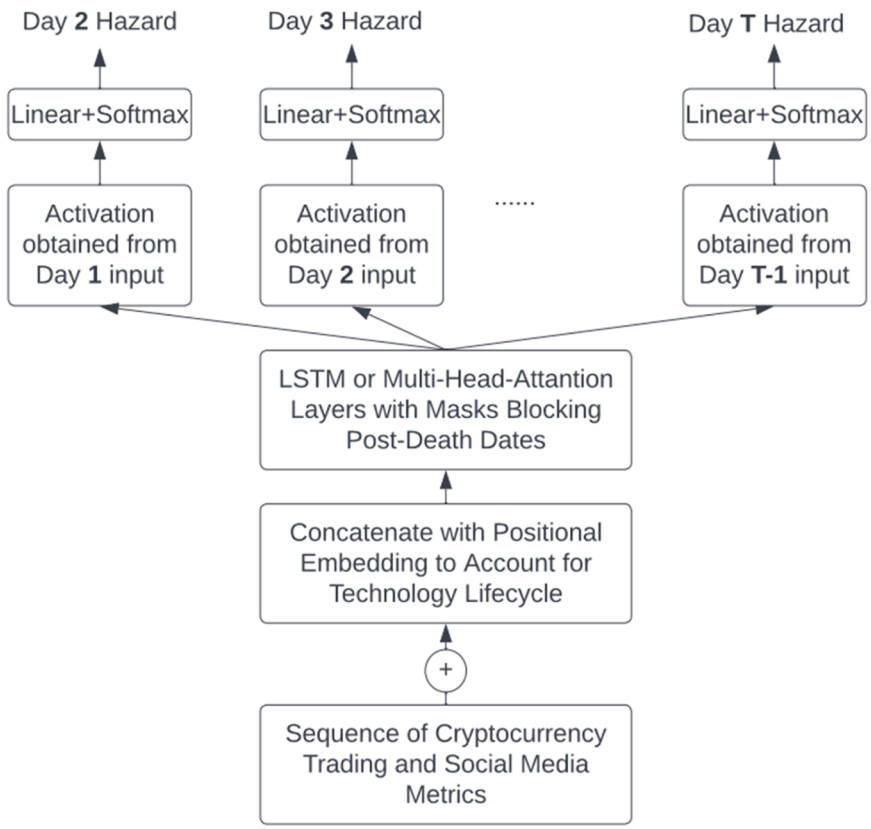


Fig. 2. LSTM/Transformer Survival Model Illustration.

After training the model, we apply it to the test set to predict the time $t+1$ hazard rate using time t covariates. The first survival analysis evaluation method we adopt is the one commonly used in the bankruptcy prediction literature [5,19]. Specifically, during each day with at least one death event, we arrange the cryptocurrencies at risk (i.e., those cryptocurrencies that "die" during this day and those that survive through it) into 10 deciles based on their predicted hazard. We then place cryptocurrencies with high predicted death hazards into the first few deciles and cryptocurrencies with low predicted death hazards into the last few deciles. We then count and aggregate the cryptocurrencies in each decile that actually experience a death event during $t+1$ through all days in the test set. If the model predicts well, this count should be very high in the top few deciles and very low in the bottom few deciles.

Table 3 presents the prediction evaluation results for both the Cox model and the deep learning survival models. Column (1) shows that 65.4 % of cryptocurrencies that experienced a death event the next day are placed in the top deciles by the Cox model because of their high predicted death hazard, indicating the general effectiveness in death hazard prediction. Column (2) shows that the LSTM survival model slightly outperforms the Cox model in prediction performance by placing 69.6 % dying cryptocurrencies in the top decile. Column (3) shows that the transformer survival model significantly outperformed both by ranking 97.8 % of dying cryptocurrencies in the top decile.

Table 3
Cryptocurrency Death Prediction Evaluation Using 10-Decile Method.

	Cox Model Trained With Both 30-day Moving Avg. and Cumulative Variables	LSTM	Transformer
	Percentage of the 186 death events in the test set in each decile		
Decile	(1)	(2)	(3)
1	65.4 %	69.6 %	97.8 %
2	13.5 %	14.7 %	0.5 %
3	7.0 %	4.9 %	0 %
4	3.2 %	2.7 %	0 %
5	4.9 %	1.6 %	0 %
6	1.6 %	2.2 %	0.5 %
7	1.1 %	1.6 %	0 %
8	1.1 %	1.1 %	0 %
9	0.5 %	0.5 %	0 %
10	1.6 %	1.1 %	1.1 %

We also use the AUROC (area under the receiver operating characteristic curve) to evaluate the models' predictive performance. Specifically, we compute the AUROC values for each day with at least one "death" event and present the summary statistics across all event days in Table 4. The result shows that the Cox model and the LSTM survival model attain comparable AUROC around 0.9, while the transformer survival model significantly outperforms both by achieving a much higher AUROC of 0.987. Both evaluation method confirms that the transformer survival model is superior in death prediction.

The high predictive performance of these models is rooted in the drastically abnormal cryptocurrency trading patterns before its "death," which can be easily picked up by the model. Fig. 3 compares the average trends of key covariates for surviving and dying cryptocurrencies over the 180 days prior to their censoring or death event. It shows that trading volume and market capitalization of dead cryptocurrencies drop sharply just before death, while those of the surviving cryptocurrencies remain largely stable. Also, for dying cryptocurrencies, the number of days since the last social media message and the price volatility also rise more than the surviving ones before their death.

6. Interpreting the deep learning survival models

In Section 4.2, we conducted a Cox regression analysis and interpreted the estimated model to derive managerial insights. However, because Cox regression relies on an assumed stochastic process where

Table 4
Cryptocurrency Death Prediction Evaluation Using AUROC.

	Cox Model Trained With Both 30-Day Moving Avg. and Cumulative Variables	LSTM-based Survival Model	Transformer-based Survival Model
Mean AUROC	0.892	0.906	0.987
Max AUROC	1	1	1
Min AUROC	0.506	0.406	0.01
Std.Dev	0.125	0.121	0.090

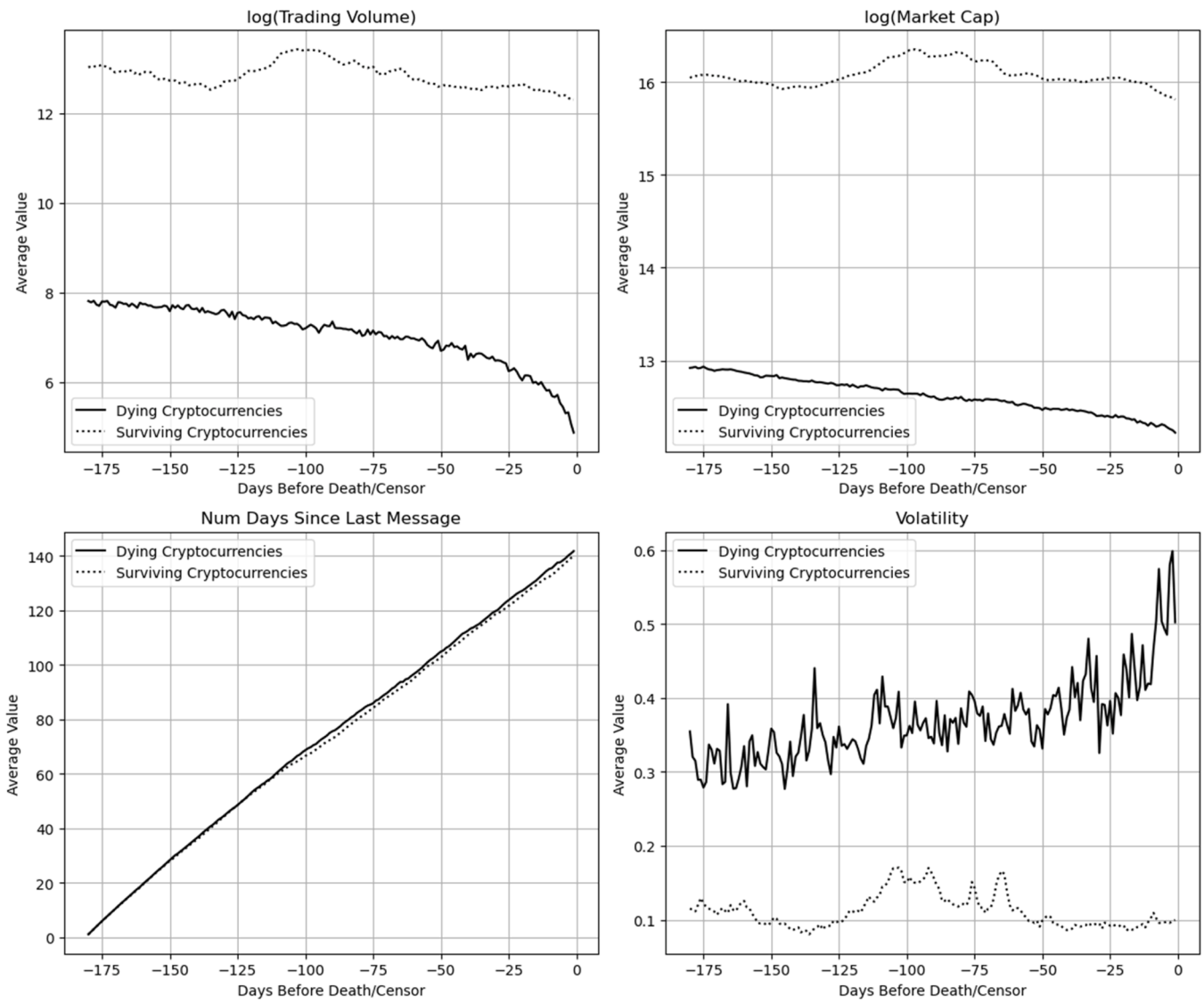


Fig. 3. Trend Near Death/Censor.

the time $t+1$ hazard is the logarithm of a linear combination of time t covariates, its interpretability is not flexible, leading to limitations in the types of queries it can address. For instance, how does a shock to one of the covariates predict the hazard rate change over time? Does this influence diminish or intensify? Moreover, what are the more subtle non-linear effects of the covariates? In this section, we examine two ways to extend the interpretability of survival analysis: the lagged effect of the covariates and the non-linear effect of the covariates.

6.1. Lagged effect

Inspired by the impulse response techniques in time series analysis, we propose a post-hoc interpretability algorithm to understand the decision-making process of the deep learning survival models and to facilitate more flexible model interpretations.

The impulse response technique is often used in vector autoregressive (VAR) systems to interpret the time series models. We will first illustrate the fundamental reasons why a VAR system is interpretable via IRF. Suppose a VAR system has p endogenous variables denoted as y_t. Further assume a lag order of 1. Then the VAR system can be written as:

$$y_t = \Pi y_{t-1} + \varepsilon_t, \tag{6}$$

where both y_t and ε_t are $p \times 1$ vectors, and Π is a $p \times p$ coefficient vector. Suppose a shock to the j-th endogenous variable occurs at time t (i.e., there is a small change to $\varepsilon_{j,t}$); then, the impact of this shock on all endogenous variables at a future point $t+h$ is simply the derivative of y_{t+h} with respect to $\varepsilon_{j,t}$, as Eq. (7) shows:

$$\frac{\partial y_{t+h}}{\partial \varepsilon_{j,t}} = \frac{\partial}{\partial \varepsilon_{j,t}}(\Pi y_{t+h-1} + \varepsilon_{t+h-1}) = \frac{\partial}{\partial \varepsilon_{j,t}}\left(\Pi^h y_t + \sum_{i=1}^{h} \Pi^{i-1}\varepsilon_{t+h-i}\right)$$
$$= \frac{\partial}{\partial \varepsilon_{j,t}}\Pi^{h-1}\varepsilon_t, \tag{7}$$

where the second equality follows from iterative substitutions using Eq. (6) and the third equality follows from the elimination of all y terms and ε terms not observed at t during the differentiation. The IRF is simply the j-th column in the Π^{h-1} matrix. Thus, the IRF derived for the VAR system only depends on the coefficient estimates Π and is independent of any intermediate y quantities from t to $t+h$. This is a desirable property because the interpretation of the model does not depend on the realization of the intermediate variables and thus is easily generalizable. For example, when assessing the impact of a shock to a variable on another variable h days later, the only information needed is the estimated coefficients while the realization of all intermediate variables or input variables from t to $t+h$ do not matter. Such interpretation may be an oversimplification, but it is easy to implement and understand.

However, when modeling a time series using deep learning models, the same IRF will depend not only on the coefficient estimates but also on all the inputs over time, as the functions involved in the computation are typically complex and non-linear, and differentiation will not entirely eliminate the intermediate inputs. First, take the LSTM survival model as an example. Suppose y_{t+1} is the predicted hazard rate during day $t+1$, x_t is the day t covariates, and m_{t-1} is the memory state of day $t-1$. Then $y_{t+1} = LSTM_{Output}(x_t, m_{t-1})$, and $m_{t-1} = LSTM_{Memory}(x_{t-1}, m_{t-2})$[2]. If a shock occurs to x_t, the IRF h days later can be written as:

$$\frac{\partial y_{t+h}}{\partial x_t} = \frac{\partial}{\partial x_t}\left(LSTM_{Output}(x_{t+h-1}, m_{t+h-2})\right)$$
$$= \frac{\partial}{\partial x_t}\left(LSTM_{Output}\left(x_{t+h-1},\ LSTM_{Memory}(x_{t+h-2}, m_{t+h-3})\right)\right)$$
$$= \ldots\cdots$$
$$= \frac{\partial}{\partial x_t}\left(LSTM_{Output}\left(x_{t+h-1},\ \ldots LSTM_{Memory}\left(x_{t+1}, LSTM_{Memory}(x_t, m_{init})\right)\ldots\right)\right) \tag{8}$$

The second equation holds by substituting m_{t+h-2} by $LSTM_{Memory}(x_{t+h-2},\ m_{t+h-3})$, and m_{t+h-3} in the expression can be substituted further by $LSTM_{Memory}(x_{t+h-3}, m_{t+h-4})$. This substitution process continues until the initial memory state m_{init} is reached in the first period. As Eq. (8) suggests, the IRF is determined by a series of nested $LSTM_{Memory}$ computations. Due to the non-linear nature of LSTM computations, taking derivatives with respect to x_t do not eliminate any intermediate inputs from x_{t+1} *to* x_{t+h-1}[3]. Eventually, we must use the chain rule to back-propagate to the gradient at x_t, which will be a function of the entire sequence x_t *to* x_{t+h-1}. Thus, the IRF can be expressed in the following form:

$$\frac{\partial y_{t+h}}{\partial x_t} = \frac{\partial}{\partial x_t} g(x_t, \ldots,\ x_{t+h-1} \mid \beta), \tag{9}$$

where $g(\cdot)$ represents the nested LSTM functions in Eq. (8) and β represents the estimated parameters in the neural network. Because the IRF depends not only on the coefficient estimates but also on the entire input sequence $x_t, \ldots,\ x_{t+h-1}$, making traditional interpretations such as "if x_t increases by ε, the hazard will increase by a certain amount" is not possible. However, if we know the joint probability distribution of $x_t, \ldots,\ x_{t+h-1}$, such as $f(x_t, \ldots,\ x_{t+h-1})$, we can compute the expectation of the differentiation as:

$$E\left(\frac{\partial y_{t+h}}{\partial x_t}\right) = \frac{\int \left[g(x_t + \varepsilon, \ldots,\ x_{t+h-1} \mid \beta) - g(x_t, \ldots,\ x_{t+h-1} \mid \beta)\right] \times f(x_t, \ldots,\ x_{t+h-1}) dx_t \ldots dx_{t+h-1}}{\varepsilon} \tag{10}$$

To calculate the expectation, we use the observed time series of covariates for all cryptocurrencies to discretely approximate the joint distribution $f(x_t, \ldots,\ x_{t+h-1})$ and then to empirically compute the value of Eq. (10). The time series of the covariates for a single cryptocurrency represents a single random draw from the joint distribution $f(x_t, \ldots,\ x_{t+h-1})$. We then compute the expected IRF by averaging the effect of the shock on the hazard across multiple draws (i.e., the full sample).

The transformer network is in a similar situation. Suppose there is a series of covariates observed from time t to $t+h$, denoted as $x_t, \ldots,\ x_{t+h-1}$, and a simple self-attention layer is applied to compute the hazard over time, denoted as $y_{t+1}, \ldots y_{t+h}$. Let $X = [x_t, \ldots x_{t+h-1}]$, $Y = [y_{t+1}, \ldots y_{t+h}]$. Also, the key, query, and value vectors in the self-attention layer are represented by XK, XQ, and XV, where K, Q, and V are the trainable parameters. Then, according to the classic self-attention procedure, $Y = softmax\left(XQ(XK)^T\right)XV$, and the IRFs are the upper portion of this function's Jacobian matrix. Only the upper portion is needed because a hazard rate can only be influenced by the covariates observed in the past. Specifically, the IRFs can be written as:

$$IRF = \begin{bmatrix} \frac{\partial y_{t+h}}{\partial x_{t+h}} & \cdots & \frac{\partial y_{t+h}}{\partial x_t} \\ \vdots & \ddots & \vdots \\ NA & \ldots & \frac{\partial y_t}{\partial x_t} \end{bmatrix} \tag{11}$$

In particular, the IRF h days later after day t is:

$$\frac{\partial y_{t+h}}{\partial x_t} = \frac{\partial}{\partial x_t} softmax\left[row_h\left(XQ(XK)^T\right)\right] XV, \tag{12}$$

where the notation row_h refers to the h-th row of a matrix. Because the softmax transformation involves taking the natural exponential operations for each value in the vector $row_h\left(XQ(XK)^T\right)$, taking derivatives with respect to x_t does not remove intermediate observations such as $x_{t+1}, \ldots, x_{t+h-1}$, again making traditional interpretations challenging.

But similarly, the expression $softmax\left[row_h\left(XQ(XK)^T\right)\right] XV$ is just a function of $x_t, \ldots x_{t+h-1}$. If we denote $softmax\left[row_h\left(XQ(XK)^T\right)\right] XV = g(x_t, \ldots,\ x_{t+h-1} \mid \beta)$, where β represents the trainable parameters Q, K, and V, we can also compute the expectation of the IRF using Eq. (10) once we know the joint probability distribution $f(x_t, \ldots,\ x_{t+h-1})$.

Table 5 illustrates the procedure to compute the expected IRF of lag l given a perturbation to covariate x. Specifically, if a perturbation ε_s occurs during day s, for each cryptocurrency i in the sample, we replace $x_{i,s}$ with $x_{i,s} + \varepsilon_s$. Then, we pass $x_{i,s}$ and $x_{i,s} + \varepsilon_s$ to the trained survival model separately to obtain two hazard rates during day $s+l$; one is the baseline death hazard without the perturbation and the other is the counterfactual death hazard after the introduced perturbation. For each lag l, we calculate the average percentage change from the baseline

[2] Assume $LSTM_{Output}(\cdot)$ and $LSTM_{Memory}(\cdot)$ are the nonlinear computations in an LSTM block calculating the output and the memory state, respectively.

[3] The nonlinear computations in a typical LSTM block consist of tanh and sigmoid functions, which involve natural exponential functions e^x. Taking derivatives with respective to e^x preserves all its input terms.

Table 5
Expected IRF Computation Procedure.

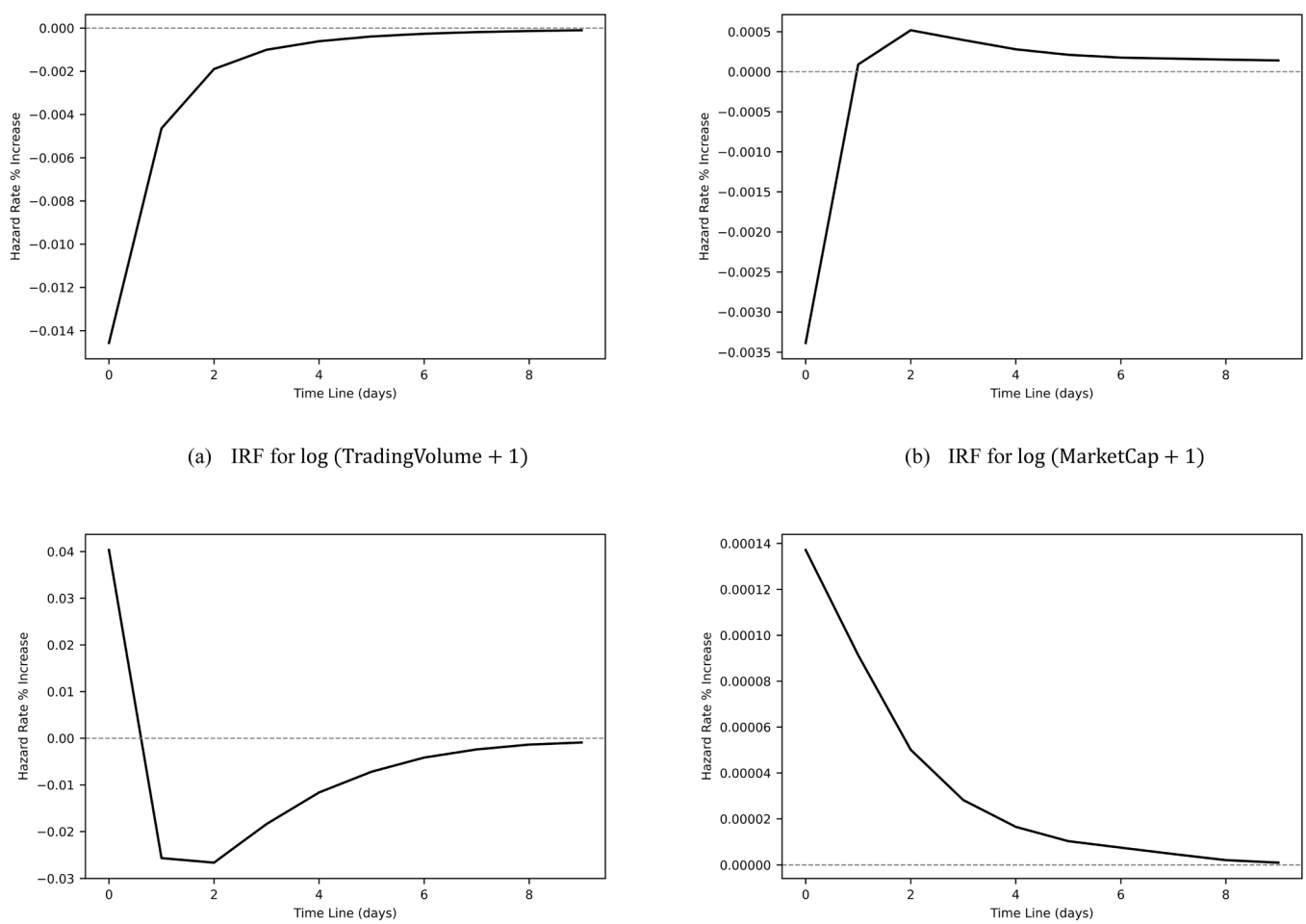
Algorithm parameter:
The max lag of the IRF L,
The date of the introduced perturbation s,
The perturbation ε_s,
Input:
The estimated LSTM or Transformer survival model $f(\cdot)$.
The sample $x_{i,t}$, i indexes the cryptocurrencies and t indexes the day.
Loop $l \in \{1, 2, \ldots, L\}$:
Initialize $I \leftarrow \varnothing$, used to collect the empirical responses due to the perturbation
Loop each sample i:
$hazard_{s+l} \leftarrow f(x_{i,s})$
$hazard'_{s+l} \leftarrow f(x_{i,s} + \varepsilon_s)$
$I \leftarrow I \cup \{hazard'_{s+l} - hazard_{s+l}\}$,
Expected Response for lag $l \leftarrow$ mean(I)
Output the expected response for each lag l.

hazard for each covariate due to the perturbation and then infer the IRF.

To test the four hypotheses using the interpretability algorithm, we compute the expected IRF for the following covariate: log(TradingVolume + 1), log(MarketCap + 1), NumDaysSinceLastMention, and Volatility. The introduced perturbation ε_s to these variables is 0.1σ, where σ is the standard deviation of the normalized covariate during day s. We observe the IRF for a maximum lag of 10 days.

Fig. 4 shows the expected IRF computed for the LSTM survival model. The expected IRF provides consistent insights with Cox regression models. First, a 10 % standard deviation increase in log (TradingVolume + 1) predicts a 1.46 % decrease in the next day's death hazard, supporting *Hypothesis 1*. This effect quickly decays to zero within 6 days. Second, a 10 % standard deviation increase in log(MarketCap + 1) predicts a 0.34 % decrease in the next day's death hazard, and the influence diminishes to zero quickly after a slight reversal, supporting *Hypothesis 2*. Third, a 10 % standard deviation increase in NumDaysSinceLastMention predicts increased death hazards during the next few days, but also with a reversal. In particular, the next day's death hazard increase due to this perturbation is 4.03 %, supporting *Hypothesis 3*. Market reversals are common situations when overactions to new information (e.g., the introduced perturbation) are subsequently corrected toward the fundamentals [69]. Overreactions and reversals exists in various types of markets such as the stock market [70], the foreign exchange market [71], and the bond market [72]. Last, a 10 % standard deviation increase in volatility predicts a 0.01 % increase in the next day death hazards and this effect decays to zero within about 8 days, supporting *Hypothesis 4*. However, this effect is comparatively less economically significant.

Fig. 5 shows the expected IRF computed for the transformer survival model, which provides similar insights that higher trading volume, market capitalization, and social media attention are associated with lower death hazard, while increased price volatility corresponds to a higher death hazard. However, one notable difference is that the magnitude of the lagged effects and the reversal effect identified by the transformer survival model is much smaller than that of the LSTM



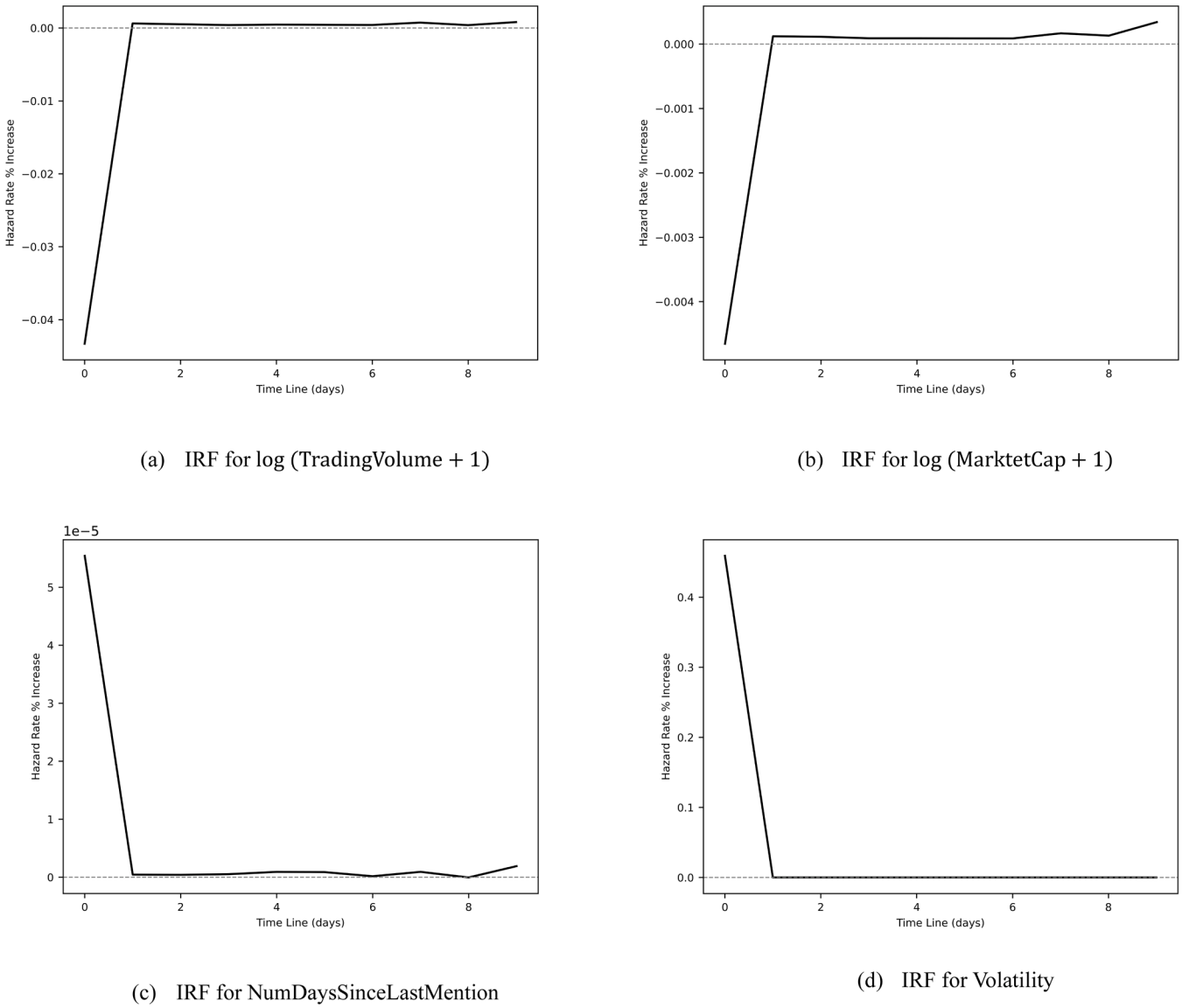
(a) IRF for log (TradingVolume + 1)

(b) IRF for log (MarketCap + 1)

(c) IRF for NumDaysSinceLastMention

(d) IRF for Volatility

Fig. 4. Expected Death Hazard IRF for LSTM Survival Model: Influences of 10 % Standard Deviation Increases.



(a) IRF for log (TradingVolume + 1)

(b) IRF for log (MarktetCap + 1)

(c) IRF for NumDaysSinceLastMention

(d) IRF for Volatility

Fig. 5. Expected Death Hazard IRF for Transformer Survival Model: Influences of 10 % Standard Deviation Increases.

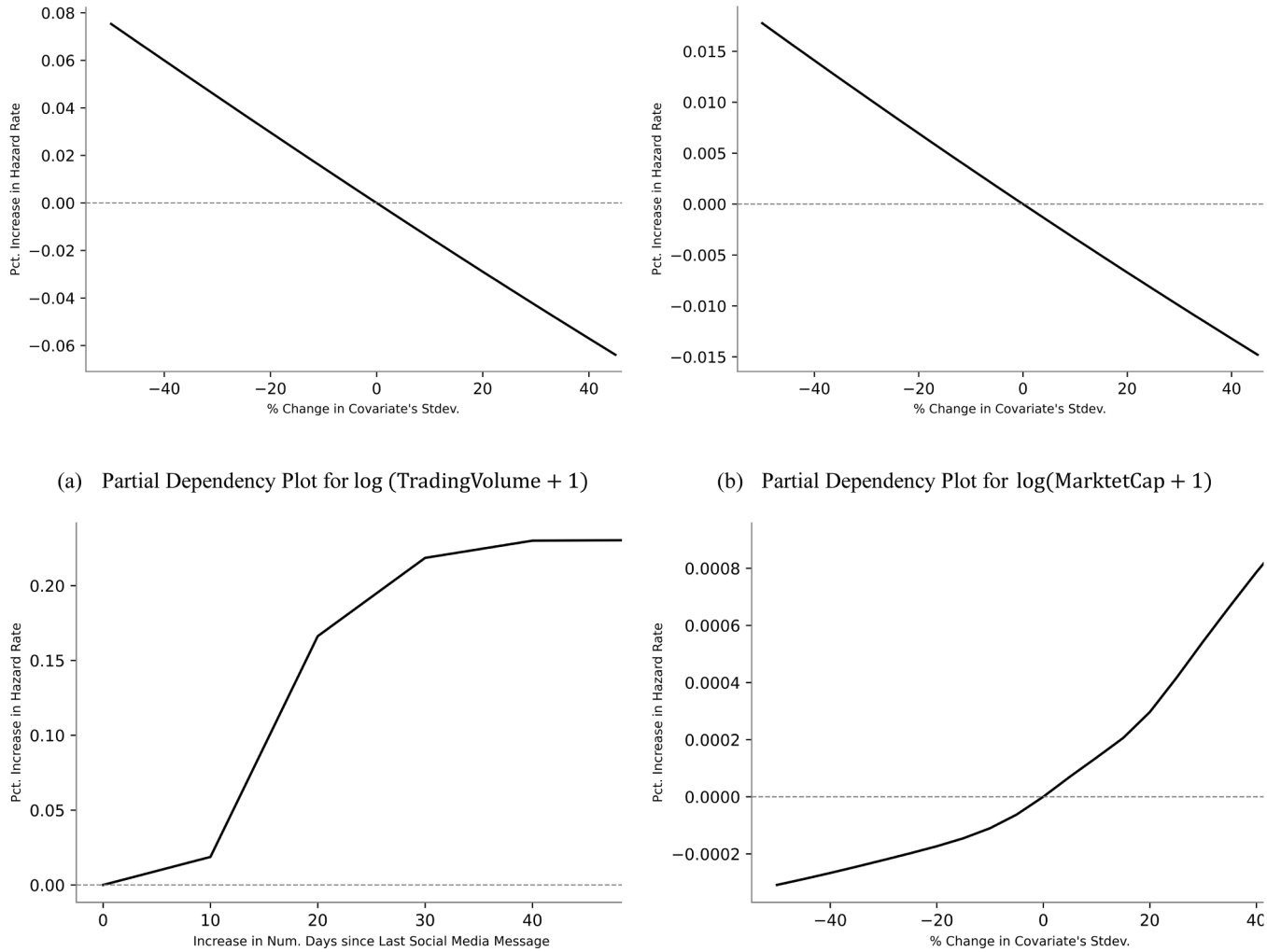
survival model, indicating a strong Markov property with respect to the covariates that the future death hazard depends primarily on the most current information, with minimal influence from the past. This interpretation aligns with the Efficient Market Hypothesis that future market dynamics depend only on new information but not on the history [73].

6.2. Non-linear effect of covariates

Cox survival model falls in the category of generalized linear models (GLMs), which allows a linear model to be related to a dependent variable via a link function. GLMs are generally interpretable because they provide estimated coefficients that indicate the effect of each covariate. However, due to their relatively simple structures, GLMs may not capture complex relationships. Their interpretability is often limited by their linear combination of covariates and the pre-defined nonlinear link function. As a result, more subtle non-linear effects of covariates might not be fully captured. For example, a relationship that appears linear in a GLM might actually be better represented by a convex, concave, or even bell-shaped curve.

Deep learning models are known for their capability to capture complex nonlinear effects [9]. In this study, we employ partial dependency plots (PDP) to evaluate the nonlinear effect of covariates on cryptocurrencies' death hazard. PDP is a model-agnostic method often used to show the marginal effect of one variable on the predicted outcome of a machine-learning model [74]. To calculate the PDP, we individually introduce a series of perturbations to each covariate of interest and observe the resulting changes in the average hazard rate over the following day. The perturbations introduced to the continuous variables, log(TradingVolume + 1), log(MarketCap + 1), and price volatility range evenly from -0.5σ to 0.5σ, where σ is the standard deviation of the normalized covariates, while the changes introduced to NumDaysSinceLastMention range from 0 to 60 days. The PDPs obtained using the LSTM and the transformer survival models are presented in Figs. 6 and 7, respectively.

The PDPs provide new yet consistent insights. For example, both the LSTM and the transformer survival models demonstrate that log(TradingVolume + 1) and log(MarketCap + 1) have a convex decreasing relationship with the next-day hazard, NumDaysSinceLastMention volatility has a concave increasing relationship with the next-day hazard, while price volatility has a convex increasing relationship with the next-day hazard. These results not only provide further support for all four hypotheses but also offer additional insights. For example, the convex decreasing relationship between trading volume and the next-day hazard implies that an increase in trading volume predicts a



(a) Partial Dependency Plot for log (TradingVolume + 1)

(b) Partial Dependency Plot for log(MarktetCap + 1)

(c) Partial Dependency Plot for NumDaysSinceLastMention

(d) Partial Dependency Plot for Volatility

Fig. 6. Partial Dependency Plot for LSTM Survival Model.

marginally diminishing negative effect on the next-day hazard.

7. Conclusion

7.1. Summary of findings, theoretical contribution, and practical contribution

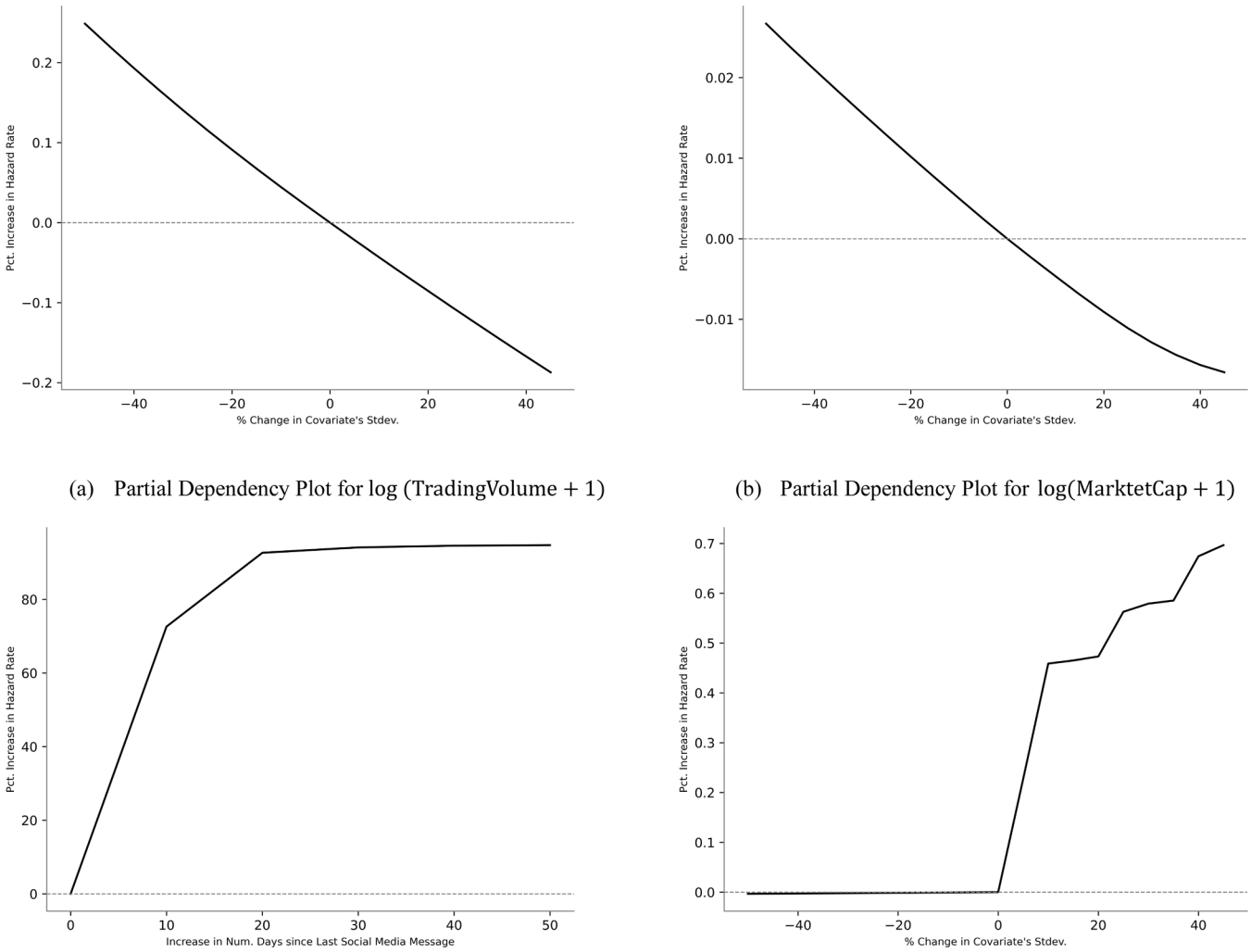
This research investigates tradable assets' "natural death" using time-dependent Cox regression and two proposed deep-learning survival models. We also designed a post-hoc interpretability algorithm to help explain the deep learning survival models. Our results suggest that a negative shock to social media attention, trading volume, and market capitalization, and a positive shock to price volatility, predicts a higher risk of cryptocurrency "death" in the following days.

Our research offers valuable implications for research by demonstrating a method to obtain interpretable insights from deep learning models for hypothesis testing using post-hoc interpretability algorithms and partial dependence plots. These interpretability approaches reveal more realistic, versatile, and detailed relationships between covariates than linear models. In the context of longitudinal data analysis, such as this study, this methodology can be used to identify both the lagged effect and the nonlinear effect of the covariates.

In the finance literature, there have been efforts to understand the termination of tradable assets such as stock delistings [18,38,39]. However, these failure events are governed by predetermined regulations [1], so that the "natural death" is not observable. Furthermore, there isn't a theory that explains how the implicit hazard of tradable assets might change over time. Our study is the first to investigate the "natural death" of tradable assets under market equilibrium, free from enforced trading termination orders. We show how the trading activity of an asset may gradually come to a stop even without enforced termination. Our study is also the first to use the cryptocurrency market to theorize the relationship between the tradable asset hazard rate and several important factors such as social media attention, trading volume, and market capitalization. The identification of such a relationship was not possible in previous literature because delisting events are deterministic in the stock market due to regulations.

This research also contributes to the survival modeling literature by identifying important drawbacks of the conventional Cox regressions and introducing a framework to develop deep learning survival models that could outperform Cox regressions in "death" prediction and in analyzing the lagged and nonlinear effects of the covariates.

Our findings offer significant practical implications as well. For example, institutional investors in cryptocurrency markets can leverage the study's results to set up alerts for cryptocurrencies at risk of failure. By closely monitoring the hazard rate changes for each cryptocurrency



(a) Partial Dependency Plot for log (TradingVolume + 1)

(b) Partial Dependency Plot for log(MarktetCap + 1)

(c) Partial Dependency Plot for NumDaysSinceLastMention

(d) Partial Dependency Plot for Volatility

Fig. 7. Partial Dependency Plot for Transformer Survival Model.

in their portfolio, they can make timely adjustments to their investments to avoid investment loss. Moreover, financial institutions could develop risk profiles or investment ratings for cryptocurrencies based on their hazard rates over time. These indicators would assist in identifying cryptocurrencies that are worth investing in or in selecting cryptocurrencies to invest for portfolios with varying levels of risk tolerance.

7.2. Limitations and future research opportunities

We recognize two main limitations in this study and suggest corresponding areas for future research. First, with the archival data we collected, the models do not establish causality. Future studies could use the same cryptocurrency market context to explore causal relationships, providing more actionable business insights. Second, some cryptocurrencies have "resurrected" after their initial failure. While our study only focuses on the first "death," investigating these "resurrection" events could also be valuable in identifying profitable investment opportunities. Future research could explore research questions such as which types of cryptocurrencies are likely to resurrect and what leads to such resurrection events.

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CRediT authorship contribution statement

Peng Xie: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nan Li:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Hongwei Du:** Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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